



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI, PILANI CAMPUS

First Semester 2026-27
Course Handout – Part II

Date: August 1, 2026

In addition to Part I (General Handout for all courses) printed on page 1 of the timetable book, this portion gives further specific details regarding the course.

Course Number : BIO F101

Course Title : INTRODUCTION TO BIOLOGICAL SCIENCES

Instructor-in-Charge : SHASHI PRAKASH SINGH (email: Shashi.singh@pilani.bits-pilani.ac.in)

Lecture Instructors : Shashi Prakash Singh, Syamantak Majumder and Rajdeep Chowdhury

Lab coordinator : MUKUL JOSHI (email: mukul.joshi@pilani.bits-pilani.ac.in),

Lab Instructors: Mukul Joshi, Shibasish Chowdhury, Vishal Saxena, Soumitra Ghosh, Sandhya A Marathe, Jitendra Panwar, Sushmita Basu, Sudeshna Mukherjee, P. Yadukrishnan, Jigneshkumar D. Prajapati, Shilpi Garg, B Vani, Prabhat N Jha, P R Deepa, Meghana Tare,

1. Course Description:

Properties of living systems, Biological macromolecules, Basic physiological processes, Genetics, Photosynthesis, Metabolism and Cellular systems. Experiments including but not limited to Microscopy, Quantification of Biomolecules, Chlorophyll extraction, Water potential in plant tissues and Cell cycle analysis.

2. Scope and Objective of the Course:

The course is aimed to provide a broad introduction to the major principles and topics in biology. The relationship of the living organism with its environment at the molecular level is highlighted in line with modern research in biological sciences. By the end of the course, the student would have gained an overall understanding of the essential aspects of biology – both as a basic science and as a branch with wide-ranging applications in industry, medicine and human health. It is also expected that using the acquired knowledge, he/she would be able to devise/suggest strategies to solve some of the imminent problems faced by mankind and the world at large through technological solutions. Students shall also get hands-on experience on fundamental aspects of practical biology.

3. Text Book:

TB: Simon, E.J. et al. Campbell Essential Biology with Physiology (5th Edition). Noida: Pearson India Education Services Pvt. Ltd., 2016.

4. Reference Book:

RB1: Enger, E.D., Ross, F.C. and David B. Bailey. Concepts in Biology (14th edition, BITS-Pilani Custom Edition 2012). New Delhi: Tata McGraw-Hill Publishing Company Ltd., 2012.

Extra Resources: Raven, P.H. et al. Biology (9th Ed). Singapore: McGraw-Hill Publishing Company Ltd., 2012.

5. Lab Manual:

Lab Manual for Introduction to Biological Sciences (First Semester 2026-27)



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6. Course Plan:

Lect No.	Learning Objectives	Topics to be covered	Chap No.	Learning Outcomes
1	Introduction to the course and biological classification	Orientation to the course content; Nature of life, Kingdoms and properties of life.	TB: 1, RB1: 1, 20	Get the initial big picture about studying biology diversity of life around us. Various aspects of bio-engineering
2	Biomolecules	Building blocks of life	TB: 3	Structure and properties of Biological Macromolecules
3	Cell structure and function	Overview of cellular organelles (nucleus, ribosomes, endo-membrane systems & cytoskeleton), virus	TB: 4 TB: 10 (virus)	Associate organelles with their cellular functions
4-5	Bioenergetics	Biochemical Pathways – cellular respiration: three stages of generating ATPs; process of fermentation	TB: 6	Examine the steps the cells use to derive energy from carbohydrates
6	Borrowing energy from nature: photosynthetic reactions	Biochemical pathways – photosynthesis: light reactions, Calvin cycle	TB: 7	Examine the steps the plant cells employ to synthesize sugars using sunlight
7-9	Cellular reproduction	Cell cycle and mitosis; stages of mitosis; cancer; meiosis – stages and generation of genetic diversity	TB: 8	Comprehend the types of cell division and the underlying control checkpoints; compare the differing mechanics in cancer cells
10-11	Patterns of Inheritance	Mendelian Genetics – laws of heredity; extensions to Mendel's law	TB: 9, RB1: 10	Application of Mendelian laws to solve heredity problems
12-14	DNA and RNA – function	DNA replication; RNA and protein synthesis: transcription and translation; mutations	TB:10	Analyze in detail the processes associated with the central dogma of biology
15-16	Genetic regulation and the process of cloning	How and why genes are controlled; cloning plants and animals; stem cells.	TB:11	Understand the importance of regulation of gene expression
17-19	Biotechnology and its Applications	Techniques of DNA manipulation; Advanced Molecular Techniques	TB:12	Critique the use of biotechnology tools for societal benefit such as DNA Fingerprinting; Sequencing



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20-22	Unifying concept of Structure and Function in Animals	Regulating internal body environment; Human excretory system & circulatory system	TB:13, 14, 15	Learn about the perfect control mechanisms that ensure homeostasis in the human body
23-24	Nervous System	Organization of the nervous system; nerve signal transmission	TB:19	Understand how life-sustaining messages get transmitted in the human body via electrical signals
25-26	Body's defense strategies	Innate and adaptive immunity; lymphatic system.	TB:16	Compare the types of defense mechanisms operating in the body when faced by diseases
27	Hormonal system	Different hormones and their modes of action	TB: 17	Understand and appreciate how life-sustaining messages get transmitted in the human body via chemical signals
28	Human reproduction and embryonic development	Human reproduction, sex and sexuality - gametogenesis; male and female reproductive systems – hormonal controls; pregnancy and early human development	TB: 18, RB1: 27	Develop a scientific awareness of reproductive health

Note: A few topics may be given for self-study, if deemed appropriate by the instructor team.

Experiment List:

Exp. No.	Learning Objectives	Topics to be covered	Learning Outcomes
Exp-1 (Week 1)	Microscopy Lab: Observing characteristic features of important specimen	Handling and use of microscopes, making observations using different Permanent/ temporary mounts to understand the relationship between structure and function. Preparation of cheek cell smears	Classification and characteristics of life forms
Exp-2 (Week-2)	Protein Lab: Measurement of total protein content using Lowry's method.	Proteins: the building blocks; their role in humans; methods to quantify proteins in different samples.	Macromolecules
Exp-3 (Week 3)	Enzyme Lab: Effect of lactase enzyme on milk	Enzyme structure, function, substrate, stability, lactose intolerance	Digestive system plus enzyme
Exp-4 (Week-4)	Transport Lab: Study the phenomenon of plasmolysis in onion peel.	Effect of tonicity of a solution on different cell types; osmosis & osmoregulation.	Cell organelles



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Exp-5 (Week-5)	Stomata & Transpiration Lab: Preparation of temporary mount of leaf epidermis to study the structure of stomata and Measurement of transpiration rate using Cobalt chloride	Visualizing stomata and understanding its role in gaseous exchange and controlling transpiration in plants. Transpiration rate and factor affecting it.	Cellular Respiration and photosynthesis
Exp-6 (Week 6)	DNA Lab: To extract DNA from banana and observe DNA on an agarose gel	Basic knowledge about the genetic material, DNA isolation and its visualization on an agarose gel.	DNA
Exp-7 (Week-7)	Immunology Lab: Measurement of hemoglobin in the human blood and determination of blood group and Rh status.	Blood typing; Blood group incompatibility, genetics behind blood group inheritance, hemoglobin and its importance; Measurement of hemoglobin content and determination of blood group and Rh status.	Blood grouping, antigen-antibody interactions
Exp-8 (Week 8)	Bioinformatics Lab: Understand Sequence Structure-Function relationship of protein through bio-informatics analysis	Understanding the application of bioinformatics in sequence comparison and analysis of various macromolecules like protein, RNA and DNA.	Sequence-Structure-Function relationship of nucleic acids and Proteins
Exp-9 (Week 9)	Chromatography Lab: Separation of chlorophyll pigments by paper chromatography	Basic knowledge about chromatography, Rf value, partition coefficient	Method for separation of compounds from mixture

7. Evaluation Scheme:

#	Evaluation component*	Duration	Weightage %	Date and Time	Nature of Component
1	Mid-Semester Test	90 min	25%	09/10 AN2	Closed-Book
2	Quizzes	-	15%	Periodic	Closed-Book
3	Class Participation		5%		
4	Lab (Participation, Performance and Record)	-	15%	As per time-table	-
5	Lab Viva	-	5%	-	-
6	Comprehensive Examination	3 hours	35%	14/12; AN	Closed/Open-Book

** Material in soft-copy will not be allowed in any announced/ surprise evaluation component. For the Open-Book Exam you are allowed to bring only the indicated books and your own handwritten notes. Printouts/xerox of lecture slides and class notes will NOT be allowed.*



8. Academic Conduct Policy:

It is expected that all students follow the highest standard of academic practice when participating in any evaluation component. Plagiarism material of any sort in evaluation components will be dealt with strictly and any such misconduct in this line will likely lead to the awarding of no marks for such evaluation components. Having a zero-tolerance for academic dishonesty, any case of misconduct, however minor, will be dealt with appropriately.

9. Grading Policy:

Award of grades would be guided by the histogram of marks and course average. **If a student obtains below 30% of the average of the top three performers or 40% of median (whichever is lower the student will not be awarded a valid grade, and the case will be reported as “NC”.**

10. Make-up Policy:

If a student misses the Mid-Semester or Comprehensive Examination due to a genuine reason (serious medical causes leading to **hospitalization (certified by doctor from BITS medical center), personal/family emergencies**) there exists a provision to apply for make-up. **Prior permission must be taken from the Instructor-in-Charge, before applying for make-up.** Make-up decisions will be made on a case-by-case basis and only genuine cases as determined by the Departmental team will be considered. No Make-Up will be granted for quiz and assignments.

11. Attendance Policy:

Each student is expected to attend all classes and participate in all evaluation components.

12. Instructor Consultation:

Instructor-in-charge: Saturday 10:30 to 11:30 PM

13. Course Notices:

Primarily, announcements regarding the course will be made in the lecture classes. Certain other notices will be communicated through Nalanda.

(Instructor-in-Charge, BIO F101)